👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 8 Engine Teardown & Rebuild (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Weeks 6-9+: engine project (teardown and rebuild), Ring End Gap/Land Clearance Handout, How to grind valves video, Engine Assembly 1, iCADEMY tech bits (engine quick check, rewind starter repair, valve clearance, throttle cable, crankcase sealing, chain, trimmer, blower and cut-off saw service, chaps)”)	**Level 1 piece**
Engine project (teardown/rebuild, not written in Canvas)	Teardown & Rebuild lab packet: step-by-step picture checklist
Ring End Gap / Land Clearance Handout	Notes B + Ring Gap steps in the lab packet
How to grind valves video	Video guide (link only)
Engine Assembly 1	Lab packet reflection + checklist (photo proof)
iCADEMY tech bits (engine check, rewind starter, valve clearance, chain/trimmer/blower/cut-off saw, chaps)	Slides 'Outdoor Power Equipment Service' + notes C (our own content)
Class slides	Deck: 15 slides, Words to Know first, “point to the picture” checks, recap
His worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Labeled parts tray (muffin tin) and a printed photo checklist at the bench
- Pair with a gen-ed partner; student owns specific steps
- Staff demonstrate each new tool first; student uses it with adult supervision

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part
- Cover one choice to make it 2

4. What to grade

- Engine project: grade on the checklist steps completed with photo proof, not speed
- Troubleshooting: L1 points to fuel, spark, compression, air cards
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Engine project checklist and grading**: The project steps and rubric are done in class and not written in Canvas; our checklist is team-written.
- **Troubleshooting**: A syllabus goal with no class materials; our lessons are team-written.
- **iCADEMY tech bits**: STIHL iCADEMY is login-only; we wrote our own service content from public knowledge.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.

- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Engine Teardown & Rebuild (Level 1)

Small Engines Overhaul · Unit 8 · goes with the class “Engine project + Engine Assembly 1 + Ring End Gap Handout” (project points). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Engine project + Engine Assembly 1 + Ring End Gap Handout”. 10 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Model first, then the student tries with adult supervision
- Picture task card at the bench
- Parts tray with labeled cups for small parts

4. What to grade (project points)

- Did the steps with support (photo proof)
- My Check page done
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: The class engine project steps and rubric are not written in Canvas; this checklist is team-written. Specs vary by engine model; staff fill them in from the class manual.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
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- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

🔧 Engine Teardown & Rebuild Notes

Small Engines Overhaul · Unit 8 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

🧩 apart _____

🥣 tray _____

🔧 torque _____

🔍 problem _____

🖼️ B. Fill in the word. Use the pictures.

5. 🧩 I take the engine _____.

6. 🥣 Small parts go in the _____.

7. 🔧 Bolts get the right _____.

8. 🔍 I look for the _____.

Word bank: 🧩 **apart** | 🔍 **problem** | 🔧 **torque** | 🥣 **tray**

⭕ C. Circle the right picture.

9. Put it back together…

 Any way	 Backward from photos

10. The engine needs…

 Pizza	 Gas, spark, squeeze, air

11. Chainsaw safety…

 Chaps	 Shorts

🔗 D. Draw a line from the picture to the word.

	spark
	compression
	air
	fuel

📖 Engine Teardown & Rebuild: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Teardown

taking the engine apart



Parts tray

cups to hold small parts



Ring end gap

space where the ring ends meet



Valve clearance

space at the valve stem



Lapping valves

grinding the valve to seal



Torque

how tight a bolt is



Troubleshooting

finding the problem step by step



Rewind starter

the pull rope part

🔧 Engine Teardown & Rebuild

Small Engines Overhaul · Unit 8 · Level 1



Name: _____

Modified from the class “Engine project + Engine Assembly 1 + Ring End Gap Handout”.

🧺 Tools: get it. Check ☐.

- ☐ 🧰 **Socket set + wrenches**
- ☐  **Flywheel puller** with adult
- ☐ 🔧 **Torque wrench** with adult
- ☐ 🌀 **Valve spring compressor** with adult

🧺 Supplies: get it. Check ☐.

- ☐ 🥣 **Parts tray** labeled cups
- ☐  **Feeler gauges**
- ☐ 🔄 **Lapping compound**
- ☐  **New gasket set**

🔧 Do one box. Check ☐ it.

1		Plug wire off. Gas drained by an adult.	☐
2		Photo before I start.	☐
3		Take off the cover and air filter.	☐
4		Parts in tray cup 1, 2, 3…	☐
5		Adult and I take off the flywheel.	☐
6		Take off the head; look at the piston.	☐
7		Check a ring gap with an adult.	☐
8		Put it back together from my photos.	☐
9		Adult checks the bolts.	☐
10		Adult checks my work.	☐

🚧 **Under construction:** The class engine project steps and rubric are not written in Canvas; this checklist is team-written. Specs vary by engine model; staff fill them in from the class manual. *Check with the class teacher.*

🔧 Engine Teardown & Rebuild · Get Ready

Small Engines Overhaul · Unit 8 · Level 1

Name: _____

👥 My job. Circle it.

- **Mechanic** — fixes
- **Tool helper** — gets tools
- **Recorder** — writes
- **Clean-up captain** — cleans

🧰 Get the tools. Check ☐.

- ☐ Socket set + wrenches
- ☐ Flywheel puller
- ☐ Parts tray
- ☐ Feeler gauges
- ☐ Torque wrench
- ☐ Valve lapping stick + compound
- ☐ Valve spring compressor
- ☐ Camera/iPad

⚠️ Be safe!

- 🥽 Glasses on.
- 👖 Coveralls on.
- 🔧 Tools with an adult.
- 🌀 Springs can jump.
- 🏭 Start outside with an adult.
- 🧼 Clean up, wash hands.

✅ **Before we start**

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

🔧 Engine Teardown & Rebuild · My Check

Small Engines Overhaul · Unit 8 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 📸 Before
- ☐ 📸 During
- ☐ 📸 After

✅ I did it! Check ☐.

- ☐ 🥽 I wore my safety gear
- ☐ 🔧 I did the engine job
- ☐ 🧽 I cleaned up
- ☐ 🤝 I helped my group

⭕ Circle one face.

🥽 **I was safe.**	😀	🙂	😕
🔧 **I did my job.**	😀	🙂	😕
⭐ **I liked it.**	😀	🙂	😕

⭐ I learned ____. Draw or write.

✅ Engine Teardown & Rebuild Quiz

Small Engines Overhaul · Unit 8 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🧩 Taking it apart…

- 🎉 Party
- 🧩 Teardown

2. 🥣 Small parts go…

- 🥣 In the tray
- ☁️ On the floor

3. 📸 Each step I…

- 📸 Take a photo
- 😴 Nap

4. ⭕ Rings go on the…

- ⛽ Tank
- 🔩 Piston

5. 🚪 Valves must…

- 🕳️ Leak
- 🚪 Seal tight

6. 🔧 **Bolts tightened to…**

- 🔧 Torque spec
- 💪 As hard as I can

7. 🔍 **Won't start, check…**

- 🎨 Paint
- ⛽ Gas and spark

8. 👖 **Chainsaw safety pants…**

- 👖 Chaps
- 🩳 Shorts

🎬 How to Grind (Lap) Valves

Small Engines Overhaul · Unit 8 · Level 1 video guide · class points pts

Name: _____

👀 Watch for these pictures.

🚪 🔄 🧴 ✅

YouTube video linked in Canvas: https://www.youtube.com/watch?v=taAP7YoVhGo

1. The valve must…

- 🚪 **Seal tight**
-  **Leak**

2. They use…

- 🧴 **Lapping paste**
- 🍯 **Honey**

3. They turn the valve…

- 🚀 **Fast forever**
- 🔄 **Back and forth**

4. Done looks like…

- 🌈 **Rainbow**
- ⚪ **Even gray ring**

⭐ **My favorite part:** _____

🔧 Engine Teardown & Rebuild Notes · KEY

Small Engines Overhaul · Unit 8 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

🧩 apart _____

🥣 tray _____

🔧 torque _____

🔍 problem _____

🖼️ B. Fill in the word. Use the pictures.

5. 🧩 I take the engine **apart**.

6. 🥣 Small parts go in the **tray**.

7. 🔧 Bolts get the right **torque**.

8. 🔍 I look for the **problem**.

Word bank: 🧩 **apart** | 🔍 **problem** | 🔧 **torque** | 🥣 **tray**

⭕ C. Circle the right picture.

9. Put it back together…

- Any way
- **Backward from photos** ✓

10. The engine needs…

- Pizza
- **Gas, spark, squeeze, air** ✓

11. Chainsaw safety…

- **Chaps** ✓
- Shorts

🔗 D. Draw a line from the picture to the word.

Key: ⛽ → fuel · ⚡ → spark · 🗜️ → compression · 🌬️ → air

⛽	spark
⚡	compression
🗜️	air
🌬️	fuel

🔑 Key written by our team from the class teacher's materials, please check.

✅ Engine Teardown & Rebuild Quiz · KEY

Small Engines Overhaul · Unit 8 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🧩 Taking it apart… Engine project

- 🎉 Party
- **🧩 Teardown** ✓

2. 🥣 Small parts go… Engine project

- **🥣 In the tray** ✓
- ☁️ On the floor

3. 📸 Each step I… Engine project

- **📸 Take a photo** ✓
- 😴 Nap

4. ⭕ Rings go on the… Ring handout

- ⛽ Tank
- **🔩 Piston** ✓

5. 🚪 Valves must… Valve video

- 🕳️ Leak
- **🚪 Seal tight** ✓

6. 🔧 Bolts tightened to… Engine Assembly

- **Torque spec** ✅
- 💪 As hard as I can

7. 🔍 Won't start, check… Troubleshooting

- 🎨 Paint
- **⛽ Gas and spark** ✅

8. 👖 Chainsaw safety pants… Service topics

- **Chaps** ✅
- 🩳 Shorts

Answers

1-B (Teardown) · 2-A (In the tray) · 3-A (Take a photo) · 4-B (Piston) · 5-B (Seal tight) · 6-A (Torque spec) · 7-B (Gas and spark) · 8-A (Chaps)

Level 1 version of the gen ed "Engine Assembly 1 / Ring End Gap Handout". Read every question aloud. Score: __ / 8.

🔑 *Key written by our team from the class teacher's materials, please check.*

🎬 How to Grind (Lap) Valves · KEY

Small Engines Overhaul · Unit 8 · Level 1 video guide · class points pts

Name: _____

👀 Watch for these pictures.

🚪 🔄 🧴 ✅

YouTube video linked in Canvas: https://www.youtube.com/watch?v=taAP7YoVhGo

1. The valve must…

- [x] 🚪 **Seal tight**
- [ ] ⚫ **Leak**

2. They use…

- [x] 🧴 **Lapping paste**
- [ ] 🍯 **Honey**

3. They turn the valve…

- [ ] 🚀 **Fast forever**
- [x] 🔄 **Back and forth**

4. Done looks like…

- [ ] 🌈 **Rainbow**
- [x] ⚪ **Even gray ring**

⭐ **My favorite part:** _____

Goes with the class “How to grind valves (video notes)” (class points pts). Same points when complete.
🔑 Key written by our team from the class teacher's materials, please check.